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Random Forest Detection of charcoal driven vegetation loss in Ghana's Afram Plains

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This study examines long-term vegetation loss and land cover change in Ghana's Afram Plains, a key forest-savannah transition zone, using integrated remote sensing and machine learning techniques over a 34-year period (1986–2021). While national forest assessments typically attribute deforestation to cocoa farming, illegal logging, and mining, this research identifies charcoal production as the primary cause of vegetation loss in this non-cocoa region. By applying supervised classification, Normalised Difference Vegetation Index (NDVI) analysis, and Random Forest algorithms to Landsat imagery, the study documents a dramatic 75% reduction in vegetation cover from 72,647 hectares (20% of land area) in 1987 to 18,019 hectares (5%) in 2021. The results reveal complex temporal dynamics, with a median annual deforestation rate of 4,728 hectares only partially offset by natural regeneration of 2,837 hectares per year. Notably, the period of the 2017–2018 national energy crisis saw unprecedented deforestation rates of 54,953 hectares per year, directly linking energy policy challenges to accelerated vegetation loss. The methodology achieved exceptional accuracy (Out-of-Bag error rates below 0.04%), ensuring high confidence in these findings. While the Afram Plains currently experiences high deforestation pressure, the landscape also exhibits notable ecosystem resilience through natural regeneration. These findings advance understanding of deforestation dynamics in forest-savannah transition zones and emphasize the urgent need for integrated conservation strategies that regulate charcoal production activities and inform sustainable forest management policies in regions where charcoal production remains a major energy source.

Keywords Deforestation, Charcoal production, Land cover change, Remote sensing, Random Forest classification, Forest-savannah transition zone, Ghana

Deforestation and forest degradation are pressing global issues that significantly impact climate change, biodiversity, and local livelihoods. In 2023 alone, the world lost 28.3 million hectares of tree cover, with tropical regions experiencing the highest rates of vegetation loss[1]; Global Forest Watch[2]. Forest-savannah transition zones, such as Ghana's Afram Plains, are particularly vulnerable due to their ecological sensitivity and intense anthropogenic pressures[3–5]. These zones serve as critical biodiversity hotspots and carbon sinks, yet their rapid degradation threatens ecosystem services and community well-being.

Advancements in remote sensing, Geographic Information Systems (GIS), and machine learning have transformed our ability to monitor and understand vegetation dynamics in such complex landscapes[6–8]. The integration of satellite imagery, vegetation indices—particularly the Normalized Difference Vegetation Index (NDVI)—and machine learning algorithms such as Random Forest has enabled comprehensive, multi-scale assessments of land cover change[9,10].

This integrated approach is especially valuable in regions where traditional ground-based monitoring is logistically challenging[11]. The synergy among these technologies is particularly impactful. Remote sensing provides foundational data through high-resolution satellite imagery, while GIS facilitates advanced spatial analysis of vegetation patterns and their interactions with socio-economic variables[12]. NDVI contributes critical insights into vegetation health and density, thereby improving the precision of change detection. When combined with machine learning—especially Random Forest Classification—this methodology has revolutionized land cover change detection in tropical regions[13].

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Recent studies underscore the effectiveness of this integrated framework. Random Forest Classification has demonstrated accuracy improvements of up to 96% over traditional classification methods[14], with the inclusion of NDVI data further enhancing accuracy by 3.87% to 4.32%[15]. GIS-based analyses have revealed intricate spatial patterns of vegetation loss, offering deeper insights into the drivers of deforestation. Collectively, these tools provide a robust and scalable framework for monitoring vegetation dynamics, offering unprecedented clarity into environmental change processes.

In Ghana, most research on deforestation has focused on forest zones affected by cocoa farming, while limited attention has been paid to forest-savannah transition areas. The Afram Plains, located in Ghana's forest-savannah transition zone, is a unique ecosystem facing significant ecological challenges due to human activities, particularly charcoal production[16–18]. While agricultural expansion is often cited as the primary driver of deforestation in Ghana (World Cocoa Foundation[19], recent evidence suggests that energy-related activities, particularly charcoal production, may play a more significant role in certain regions.

Despite its ecological importance, the Afram Plains faces alarming rates of degradation, mainly due to human activities such as charcoal production[18]. Understanding the patterns and drivers of vegetation loss in this region is crucial for developing effective conservation strategies and sustainable resource management practices.

This study thus aimed to quantify the extent and rate of vegetation loss in the Afram Plains over a 34-year period (1986–2021) using Landsat imagery, Random Forest classification, and NDVI analysis; Identify and analyze the primary drivers of deforestation in the region; Assess the temporal and spatial patterns of land cover change; and evaluate the relationship between human activities and observed land cover changes.

To achieve these objectives, we employed an integrated approach combining remote sensing, GIS, and machine learning. Landsat images were selected for their long-term coverage and suitability for change detection. Images were paired by month to minimize seasonal and phenological variation, and only those with less than 10% cloud cover were used to ensure data quality. Pre-processing and analysis were conducted in QGIS, with supervised classification and Random Forest algorithms implemented in R. NDVI was calculated to provide a quantitative measure of vegetation health and density, further enhancing classification accuracy and change detection.

This research contributes to the growing literature on vegetation monitoring in transition zones and provides crucial insights for conservation planning and resource management in Ghana's rapidly changing landscapes.

Methodological framework

The theoretical foundation of this study is grounded in the theory of land cover change, which posits that complex interactions between human activities, natural processes, and underlying socio-economic and environmental conditions drive land cover changes[20,21]. To effectively analyze and quantify land cover changes over time in the Afram Plains, we adopt an integrated methodological framework that combines remote sensing techniques with machine learning algorithms and ground truth information. This approach allows us to accurately detect and interpret changes in land cover, even in resource-constrained settings with limited access to high-resolution data.

Land cover change analysis has evolved significantly with the incorporation of machine learning algorithms, such as the Random Forest algorithm, which has been shown to outperform traditional methods in accuracy and the ability to discern subtle variations within classes[22–24]. This study employs supervised classification techniques facilitated by the Random Forest algorithm to interpret satellite imagery and identify distinct land cover classes. Additionally, the integration of the Normalized Difference Vegetation Index (NDVI) provides a quantitative measure of vegetation health and density, further enhancing the accuracy of land cover classification[25,26].

The combined use of machine learning algorithms and vegetation indices allows for a more nuanced understanding of the complex dynamics of land cover change in the Afram Plains. This integrated methodological framework captures the interplay between human activities, natural processes, and ecological patterns[27,28], providing valuable insights that can inform policy decisions and shape priority actions related to land-use planning and ecosystem restoration.

Land Cover Change Detection is a powerful tool in remote sensing that involves analyzing changes in spectral signals of pixels to identify landscape conversions from one dominant feature to another[29]. Landsat data, with its continuous observations since the 1980s[30], has been instrumental in studying tree cover and vegetation changes[31]. However, traditional post-classification comparison methods had limitations in detecting subtle differences, especially with low-resolution images[32].

The integration of machine learning algorithms, such as Random Forest, has revolutionized land cover change detection[32]. In this study, supervised classification involves defining training sites of known vegetation and applying the algorithm to automate the process across the landscape.

The Random Forest algorithm, developed by Ned Horning (available at https://github.com/nedhorning/RandomForestForRemoteSensing), is a non-parametric algorithm that combines the strengths of multiple classification methods[33]. It is well-suited for the Afram Plains, where high-resolution images are limited, providing accurate estimates of land cover change using lower-resolution images.

The Random Forest algorithm offers several key advantages that enhance its effectiveness in various applications. These advantages include its robust performance with limited training data, ability to manage non-linear relationships, and built-in cross-validation through Out-of-Bag (OOB) error estimation. Additionally, it demonstrates resistance to overfitting and can effectively handle missing values and unbalanced datasets[34,35].

NDVI is a widely used index for assessing vegetation health and density[36]. NDVI values range from − 1 to 1, with higher values indicating denser and healthier vegetation[37,38]. By incorporating NDVI with remote sensing data, this study gains additional insights into vegetation dynamics, further enhancing the accuracy of land cover classification and change detection.

The Random Forest classifier's inbuilt error estimates (OOB error rate)[22]) were used as the primary accuracy metric. The Random Forests Classifier splits the data into training and test sets (remaining one-third-out-of-bag

samples-OOB)[39,40]. Using the output of the confusion matrix and the feature class correlation plot, the training dataset was checked for errors and reclassified. A confusion matrix is a tabular representation of Observed vs. predicted values. Iteratively refined training data until satisfactory accuracy was achieved.

Materials and methods

Study area

The study focused on Ghana's Afram Plains (Fig. 1), a 5,040 km2 (504,000 kha) forest savannah transition zone in the eastern region, selected due to reports of significant vegetation loss from charcoal production, agriculture, and fires[41]; Appiah, 2013). Afram Plains is a quasi-island community in the northernmost part of the Eastern Region. It is located at Latitude 6.912698° and Longitude of -0.141224°, covering a total land area of approximately[42] and a perimeter of 303 km (Google Earth Pro). At the south and east boundaries is the Volta River, at the west is Sekyere-East and Asante-Akim in the Ashanti region, and at the north is the Sene and Atebubu Brong Ahafo Region (Ghana Statistical Services[43].

The Afram plains has high temperatures and yearly rainfall between 1300 mm and 1400 mm (Appiah[16]. The vegetation predominantly consists of savannah interspersed with short deciduous trees[42]. The selection of Afram plains as the study area was based on reports of significant destruction of vegetation cover of the plain due to forest fires, agriculture, and charcoal production. Moreover, Afram plains fall within the savannah transition zone, known to host desirable species for charcoal production, making it a vital charcoal production area and a major source of charcoal supply to the capital city of Accra (Environmental Protection Agency[44].

Data sources and tools

To detect tree cover loss, multispectral satellite images from the Landsat 5, Landsat 7, and Landsat 8 sensors were used. Landsat 7, and 5 images for Afram Plains (Path – 193; Row – 55) were downloaded, with a specific criterion of selecting images with a cloud cover of less than 10% to ensure data quality[45].

A time series dataset of multispectral Landsat TM images covering the period 1986–2021 was obtained for this study. All images were downloaded from the US Geological Survey website (https://earthexplorer.usgs.gov/) . The pre-processing and analysis of the multispectral images for tree cover loss were conducted using the open-source GIS application platform QGIS (version 3.18.1). This software was employed to perform necessary data preparation and processing tasks.

Two measures were implemented to differentiate real change from spectral changes: (1) pairing images from different years within the same month to minimise seasonal and phenological variations, and (2) ensuring that all collected images were nearly cloud-free. The Random Forest algorithm, implemented in R software (version 4.0.4[46] was utilised for the change detection analysis. The algorithm generated the change detection map, which was further analysed using R for additional insights and interpretation.

Seventeen images were downloaded and utilised in this study. Change detection was performed for 14 pairs of dates, with considerations given to commonality in season (months) for accurate comparison and analysis.

Date selection and temporal coverage

The selection of specific years for our analysis was based on several methodological and practical considerations. While our study period spans 1986–2021, the specific years selected had the clearest, most usable Landsat images with minimum cloud cover (less than 10%). Gaps between selected years reflect periods where available

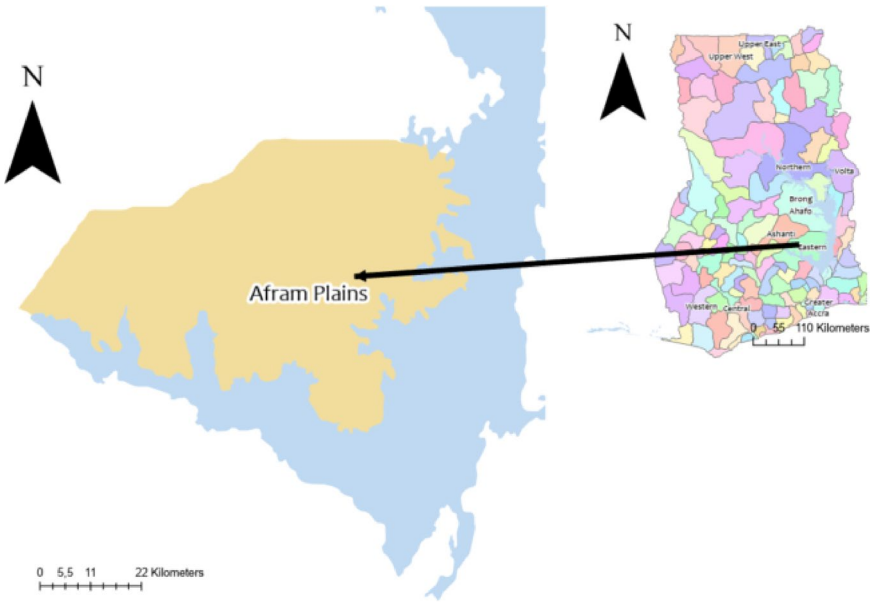


Fig. 1. Locator map of Ghana showing the Afram Plains study area; Source: Author curated, using QGIS 3.44.

imagery did not meet our quality criteria. While our change detection analysis included data from multiple dates throughout the study period (Table 2), we specifically selected February measurements for the NDVI vegetation cover estimation comparison in (Table 1), to eliminate the confounding effects of seasonal variations on NDVI values.

Research design/strategies of enquiry

This study employed a robust research design that integrated supervised classification with the Random Forest algorithm and land cover estimation using the Normalized Difference Vegetation Index (NDVI). Supervised classification, facilitated by the Random Forest algorithm, allowed for automated and accurate classification of land cover change, overcoming the limitations of traditional post-classification methods. The NDVI estimation quantitatively measured vegetation cover and health by utilising the near-infrared and red band reflectance ratios. NDVI thresholds were carefully selected and applied based on prior ground truth knowledge of actual ground features. Areas of forest were delineated using ground truth knowledge of the untarred road network that run across the entire plain. This was used to set the threshold for NDVI values of non-forest areas/bareland or built up areas and distinguishing these from degraded forest patches or temporary burnt forest areas.

The integration of supervised classification and NDVI estimation created a comprehensive land cover change analysis framework. This framework leveraged advanced remote sensing techniques and machine learning algorithms to uncover valuable insights into land cover dynamics. The Random Forest algorithm script, developed by Ned Horning (available at https://github.com/nedhorning/RandomForestForRemoteSensing), was adapted and used utilized identify distinct land cover classes and detect changes over time.

Research procedure

The research procedure involved several steps in preprocessing the images, estimating vegetation cover using NDVI, and performing supervised classification using the Random Forest algorithm to detect changes and assess accuracy.

Data acquisition

Seventeen cloud-free multispectral Landsat images (Landsat 5, 7, and 8) for the Afram plains (Path – 193; Row-55) spanning 1986–2021 were obtained from the US Geological Survey. Analysis of vegetation loss and land cover change was done over this 34-year period. For NDVI-based vegetation cover estimates, the earliest available data point is from February 1987, with subsequent comparisons made through February 2021. Images were paired by month to minimise seasonal variation[45], resulting in 14 date pairs on which the change detection analysis was performed. To ensure data quality, only data with cloud cover of less than 10 per cent was used.

Image preprocessing

Images were preprocessed in QGIS (v3.18.1) using the Semi-Automatic Classification Plugin for atmospheric correction (DOS1 algorithm)[47]. After correction, images were band stacked- individual spectral bands were merged into a single multi-band raster dataset, maintaining the appropriate band order for each Landsat sensor (e.g., Bands 5-4-3 for Landsat 5/7, Bands 6-5-4 for Landsat 8). Images were enhanced by creating colour composites and adjusting image statistics (mean ± 2 standard deviations) to enhance contrast and facilitate the identification of vegetated and unvegetated areas. Processed images were clipped to the land extent of the Afram Plains shapefile. Changes were detected for all available dates to build confidence in the change detection results.

NDVI vegetation cover estimation

The Normalized Difference Vegetation Index (NDVI) was calculated for each image using the standard formula, with thresholds set based on ground truth data to distinguish vegetation from non-vegetation.

$$\frac{(\text{Near Infrared (NIR)} - \text{Red})}{(\text{Near Infrared (NIR)} + \text{Red})} \rightarrow \text{NDVI}$$

where NIR and Red are the reflectance values in the near-infrared and red bands, respectively. NDVI values range from – 1 to 1, with higher values indicating denser and healthier vegetation.

Different NDVI thresholds were applied to classify and distinguish between various vegetation types. The selection of NDVI threshold values was guided by thorough ground survey information on the location of built-up areas and road networks (knowledge of the location of road networks and built-up areas guided the selection of the threshold for barren land). This approach ensured that changes were detected based on the analysis of

YEAR	Month	Area/ha	Percentage cover
1987	Feb	72,647	20%
2000	Feb	57,504	15%
2002	Feb	34,011	9%
2003	Feb	39,747	10%
2016	Feb	27,641	7%
2021	Feb	18,019	5%

Table 1. Change in vegetation cover (1987–2021).

vegetated non-built-up areas. The land area related to the various NDVI values was then quantified using the Raster Layer Unique values report tool of QGIS. The output file was exported to Excel for cleaning and analysis.

Supervised classification

A supervised classification approach using the Random Forest algorithm (implemented in R v4.0.4) was applied. Training data for seven land cover classes were iteratively refined, with a minimum of 10 and up to 40 trained samples (Fig. 2). Refinement and adjustment to the number of training samples were guided by the output of the confusion matrix and accuracy assessments, and were repeated for each classification period until a satisfactory accuracy was obtained. The trained samples were also cross-validated with NDVI values by overlaying the NDVI map on the atmospherically corrected image of the Afram Plains to ensure representativeness and minimise misclassification.

Change detection

Change detection was performed for 14 date pairs using the Random Forests Algorithm R Script machine learning method. Images from similar months were compared across two dates to quantify land cover transitions (deforestation, regeneration, stable classes). Post-classification filtering using the majority filter tool and sieve tools of QGIS removed isolated pixel noise. Area changes were quantified using the Raster Layer Unique values report of QGIS. The result was uploaded into Excel, cleaned up, and analysed in R. The actual changes were quantified using the Raster Layer Unique values report of QGIS. The result was uploaded into Excel, cleaned up, and analysed in R.

Accuracy assessment

Classification accuracy was evaluated using OOB error rates and confusion matrices. A new code was added to calculate the efficiency of the model. The model's efficiency also informed the optimisation of the training data until satisfactory.

$$\text{efficiency} < -\text{sum}\left(\text{diag}\left(\text{confMatrix}\right)\right)/\text{sum}\left(\text{confMatrix}\right)$$
$$\text{efficiency}$$

Area estimates were adjusted for classification error, and uncertainty was quantified using standard error and 95% confidence intervals following established protocols[48]. For example, in February 2021, the estimated vegetation area was 178,392,456 m^2 (SE: 1,346,000 m^2; 95% CI: 175,764,456–181,020,456 m^2).

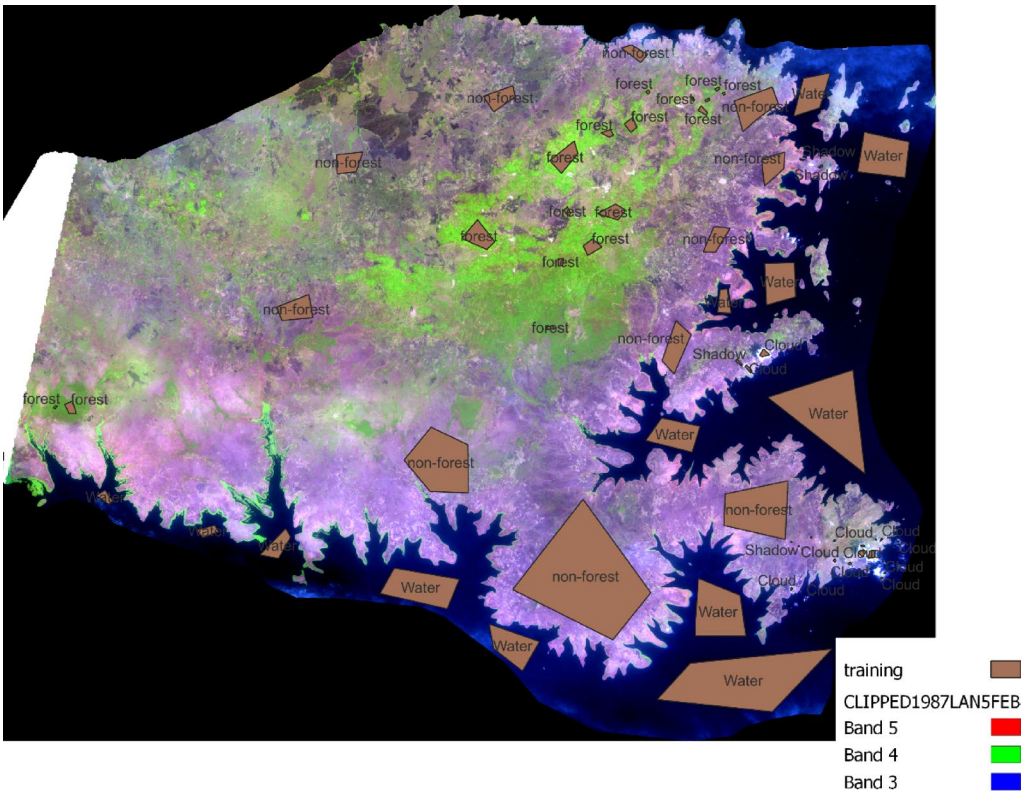


Fig. 2. Classification map showing the number of training data for Random Forest-supervised classification: Source: Author curated using QGIS 3.18.1.

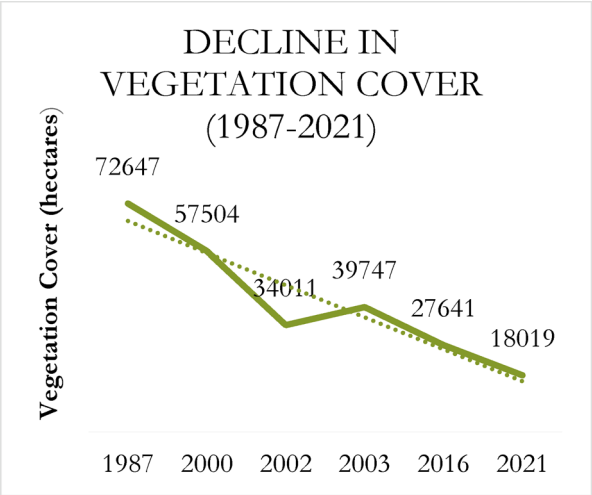


Fig. 3. Decline of vegetation cover in the Afram Plains (1987–2021); Source, Author curated using Excel for Microsoft 365.

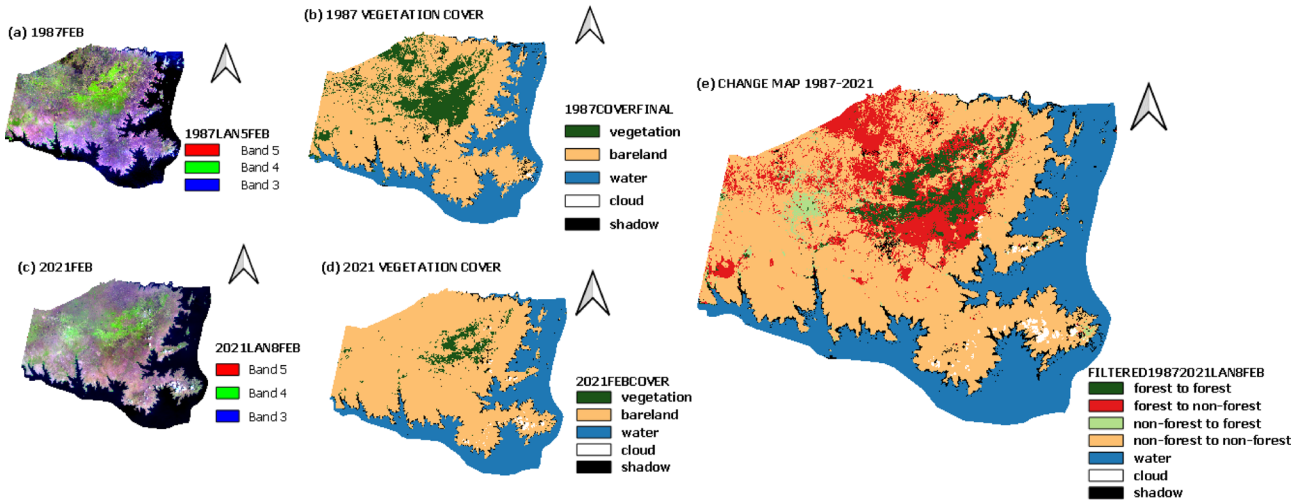


Fig. 4. Multi-temporal land cover change in the Afram Plains, Ghana (1987 to 2021) (**a**) 1987 raw satellite image; (**b**) 1987 classified vegetation cover; (**c**) 2021 raw satellite image; (**d**) 2021 classified vegetation cover (**e**) Change map 1987–2021; Source Author curated.

Results

NDVI analysis results

NDVI analysis revealed a consistent and significant decline in vegetation cover across the Afram Plains over the 34-year study period (Fig. 3). Vegetation cover in Afram plains declined from 72.6 kha (20% of total land area) in February 1987 to 18 kha (5%) in February 2021, representing a total loss of 54,628 hectares (75% of 1987 vegetation cover) (Table 1).

The mean annual rate of vegetation loss was 1,607 hectares, representing approximately 2.2% of the 1987 vegetation cover per year. However, this rate varied significantly across different periods, with the highest rates of loss occurring during 2016–2021 (1,924 ha/year).

Vegetation cover estimation and land cover change detection using supervised classification

Supervised classification analysis confirmed and expanded upon the NDVI findings, providing detailed insights into the spatial patterns of vegetation change. The classification identified four primary land cover transitions: stable forest, deforested areas, regenerated forest, and stable non-forest areas. The supervised classification results for vegetation cover estimation and land cover change detection confirmed the decline in vegetation cover in Afram Plains from February 1987 to 2021 (Fig. 4).

Temporal patterns of change

The analysis of 14 date pairs revealed distinct temporal patterns, shown in (Table 2). This table provides a comprehensive overview of forest cover changes across multiple time periods, ranging from single years to multi-decade spans between 1986 and 2021. It tracks three types of land cover transitions: Forest to Forest (stable forest), Forest to Non-Forest (deforestation), and Non-Forest to Forest (reforestation or afforestation).

The result reveals a complex dynamic of forest cover change. In most periods, the area converted from forest to non-forest exceeds the area converted from non-forest to forest, indicating a net loss of forest cover. However, the rates and magnitudes of these changes vary significantly across different timeframes.

Some periods show intense deforestation, such as 1999–2015, where 72,638 hectares were converted from forest to non-forest. Conversely, significant reforestation efforts are evident, notably in 2002–2020, where 86,321 hectares were converted from non-forest to forest.

The varying timescales in the dataset allow for analysis of both short-term fluctuations and long-term trends. Shorter periods often show more dramatic annual changes. For example, the highest short-term deforestation

Change	Period	No. of years	Area (ha)	Forest to Non-Forest/yr	Non-Forest to Forest/yr
Forest to Forest	2002–2003	1	10,192		
Forest to Non-Forest	2002–2003	1	20,146	20,146	
Non-Forest to Forest	2002–2003	1	17,160		17,160
Forest to Forest	2017–2018	1	26,927		
Forest to Non-Forest	2017–2018	1	54,953	54,953	
Non-Forest to Forest	2017–2018	1	13,995		13,995
Forest to Forest	2000–2002	2	15,307		
Forest to Non-Forest	2000–2002	2	49,097	24,549	
Non-Forest to Forest	2000–2002	2	14,395		7198
Forest to Forest	2015–2017	2	29,780		
Forest to Non-Forest	2015–2017	2	64,846	32,423	
Non-Forest to Forest	2015–2017	2	54,070		27,035
Forest to Forest	2016–2021	5	8214		
Forest to Non-Forest	2016–2021	5	45,235	9047	
Non-Forest to Forest	2016–2021	5	6142		1228
Forest to Forest	1991–1999	8	42,319		
Forest to Non-Forest	1991–1999	8	62,913	7864	
Non-Forest to Forest	1991–1999	8	28,442		3555
Forest to Forest	1987–2000	13	24,862		
Forest to Non-Forest	1987–2000	13	63,924	4917	
Non-Forest to Forest	1987–2000	13	18,554		1427
Forest to Forest	2003–2016	13	6323		
Forest to Non-Forest	2003–2016	13	29,189	2245	
Non-Forest to Forest	2003–2016	13	15,199		1169
Forest to Forest	1986–2002	16	16,505		
Forest to Non-Forest	1986–2002	16	17,531	1096	
Non-Forest to Forest	1986–2002	16	65,316		4082
Forest to Forest	1999–2015	16	25,402		
Forest to Non-Forest	1999–2015	16	72,638	4540	
Non-Forest to Forest	1999–2015	16	21,289		1331
Forest to Forest	2003–2019	16	17,416		
Forest to Non-Forest	2003–2019	16	50,763	3173	
Non-Forest to Forest	2003–2019	16	13,893		868
Forest to Forest	2002–2020	18	13,498		
Forest to Non-Forest	2002–2020	18	41,552	2308	
Non-Forest to Forest	2002–2020	18	86,321		4796
Forest to Forest	1986–2020	34	16,507		
Forest to Non-Forest	1986–2020	34	22,736	669	
Non-Forest to Forest	1986–2020	34	72,063		2119
Forest to Forest	1987–2021	34	20,896		
Forest to Non-Forest	1987–2021	34	63,734	1875	
Non-Forest to Forest	1987–2021	34	7345		216

Table 2. Change detection output for 14 date pairs (1986–2021).

rate observed was 54,953 ha/year for the 2017–2018 period, while longer periods tend to have lower annual rates. The lowest long-term deforestation rate was 669 ha/year for the 1986–2020 period (34 years). Similar to deforestation, regrowth rates show significant variability across different time periods. Faster regrowth was observed for shorter periods in 2015–2017, where 54,070 ha was revegetated, and the lowest regrowth was observed for the longest date range from 1987 to 2021, where only 216 ha was revegetated.

The distribution of Forest to Non-Forest area change, shown in a boxplot (Fig. 5) indicates significant variability in the rate of forest conversion, with a median of 4,728 hectares per year.

The distribution of Non-Forest to Forest area change, shown in a boxplot (Fig. 6), suggests moderate levels of regeneration on average (median of 2,837 hectares per year), with some periods of exceptionally high forest regrowth.

Comparing the vegetation loss and vegetation retained data (Fig. 7) shows a clear upward trend in vegetation loss over the years, with the magnitude of vegetation loss consistently exceeding the vegetation retained, indicating an overall net decline in vegetated areas.

The analysis suggests an increasing pressure on natural ecosystems and a concerning pattern of deforestation or land-use change in the Afram Plains region over the 34-year period studied.

Forest cover change was analyzed for 14 date pairs spanning 1-year to 34-year periods. Both deforestation (Forest to Non-Forest) and reforestation (Non-Forest to Forest) trends were observed.

The analysis of the "Forest to Non-Forest" rows showed high variability in deforestation rates, with some notable spikes observed in the 2017–2018 period. However, when averaged over longer periods, the long-term trend of forest-to-non-forest conversion was generally lower than the short-term peaks.

The lowest long-term deforestation rate was 669 ha/year (1986–2020), while the highest short-term rate was 54,953 ha/year (2017–2018). This suggests there are significant year-to-year fluctuations in deforestation rates.

The "Distribution of Forest to Non-Forest Area Change" (Fig. 5) shows a wide range of variability in the rate of forest conversion to non-forest areas across different periods, with a median of 4,728 hectares per year. The large MAD of 5,017 hectares compared to the median indicates high variability.

The distribution of Non-Forest to Forest area change (Fig. 6) shows a right-skewed pattern, with a longer upper whisker and outliers at the top. This suggests most periods have moderate levels of reforestation, but some periods have exceptionally high forest regrowth rates.

The median value of reforestation (Non-Forest to Forest) is 2,837 hectares per year, with a high variability indicated by the MAD of 2,689 hectares. Two notable outliers above 20,000 hectares suggest periods of extraordinary reforestation or favorable conditions for natural forest regeneration.

Comparing the median deforestation rate (4,728 ha) to the median reforestation rate (2,837 ha) suggests that, on average, more forests are lost than regained in the Afram Plains region.

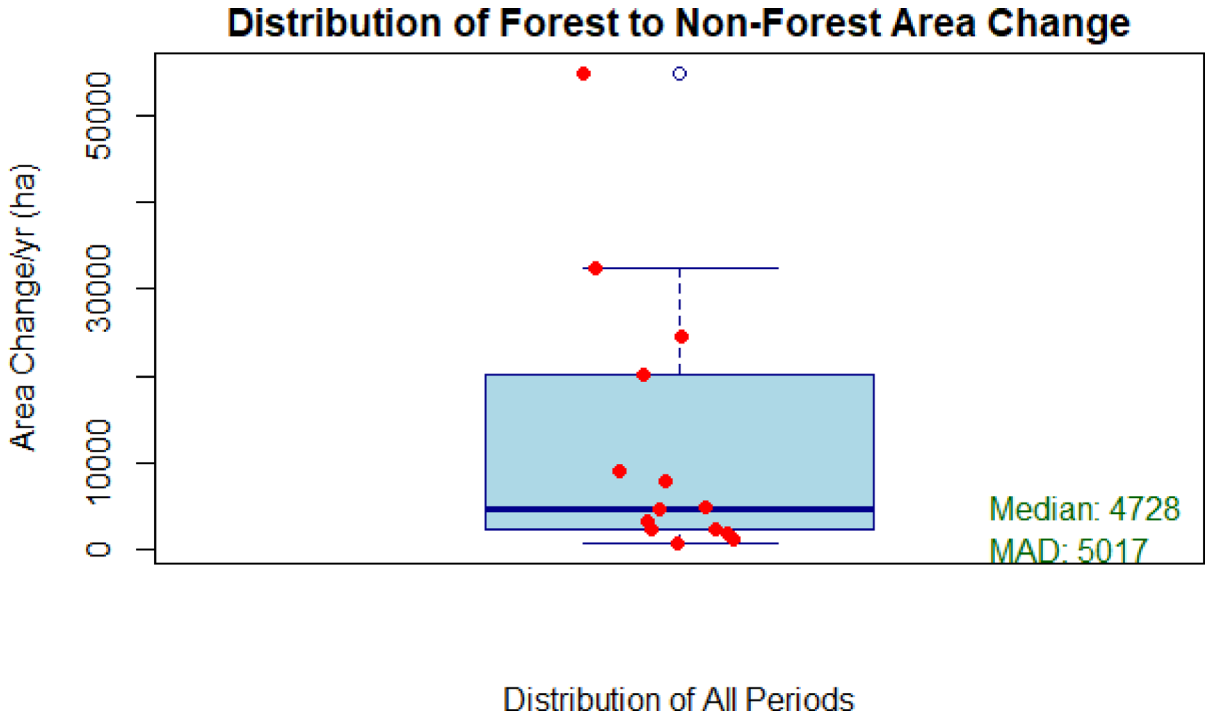


Fig. 5. Boxplot showing distribution of Vegetation Loss per year/ha across all assessment periods in the Afram Plains. Each red dot represents the annual change for a specific interval, while the box and whiskers summarize the interquartile range and variability of change rates. The median annual change is 4728 ha/yr, and the median absolute deviation (MAD) is 5017 ha/yr, indicated in green. Outliers, including a notable spike exceeding 50,000 ha/yr.

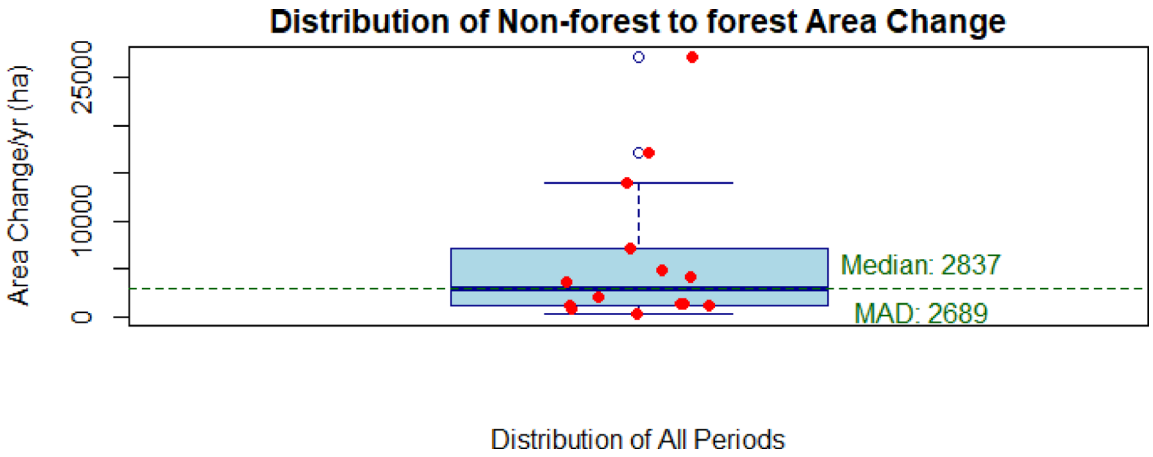


Fig. 6. Boxplot showing distribution of Regrowth year/ha for all assessment periods in the Afram Plains. Each red dot represents the annual change for a specific interval. The blue box shows the interquartile range (IQR), with the horizontal line indicating the median value (2837 ha/yr). The green dashed line marks the median, and the green text denotes the median absolute deviation (MAD = 2689 ha/yr).

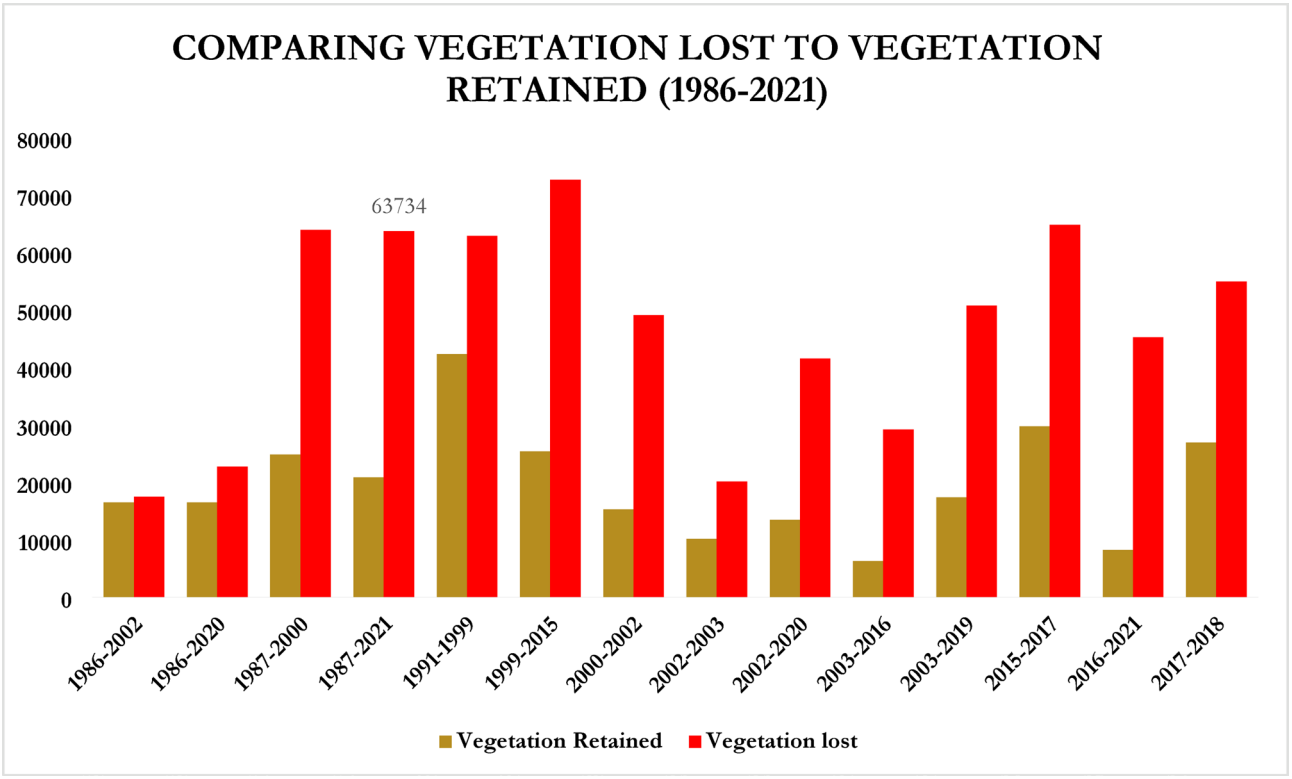


Fig. 7. Comparing vegetation lost (ha), and vegetation retained (ha) for 14 date pairs from 1986 to 2021.

This highlights the complex and variable nature of forest cover change in the Afram Plains, with both deforestation and reforestation trends observed, but with deforestation consistently exceeding reforestation on average.

Figure 7 shows a clear upward trend in vegetation loss over the years, with significant spikes observed in certain periods, such as 2015–2019. This upward trend in vegetation loss suggests an increasing pressure on natural ecosystems and a concerning pattern of deforestation or land-use change in the Afram Plains region. In contrast, the vegetation retained values exhibit more fluctuations, with some periods seeing increases in retained vegetation (e.g., 1987–2000, 2002–2009) and others experiencing declines. By directly comparing the vegetation loss and vegetation retained data, it becomes clear that the magnitude of vegetation loss consistently exceeds the

OOB error rate estimate=0.02%

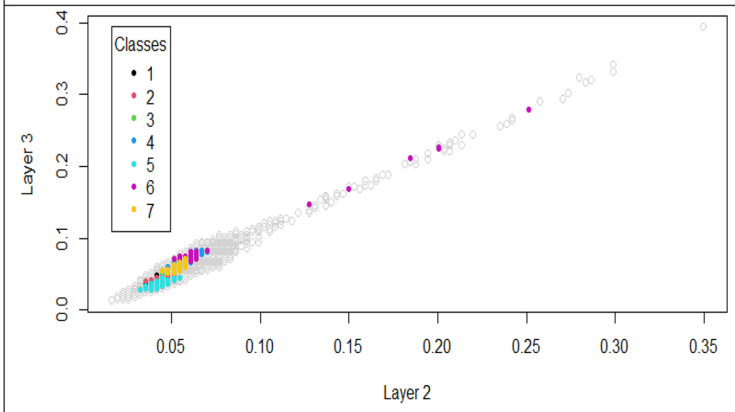
Efficiency =98.4%

> cat ("Confusion matrix\n")

Confusion matrix

> print(randfor$confusion)

	1	2	3	4	5	6	7	Class. error
1	98	2	0	0	0	0	0	0.02
2	6	92	0	0	0	1	1	0.08
3	0	0	100	0	0	0	0	0.00
4	0	0	1	99	0	0	0	0.01



Feature space plot 19872021feb change detection

Table 3. OOB error rate and feature space plot of February 1987–2021 Random Forest change detection.

vegetation retained, often by a wide margin. This imbalance between vegetation loss and vegetation retained suggests an overall net decline in vegetated areas in the Afram Plains region over the 34-year period studied.

The trend is shown in Fig. 7 paints a concerning picture of the region's natural ecosystems, with a clear and persistent pattern of vegetation loss outpacing vegetation retention, leading to an overall net decline in vegetated areas over the long term.

Accuracy of Random Forest classification

The Out-of-Bag (OOB) error rate (Table 3), which indicates the classification error rate without needing an independent validation dataset, was below 0.04% for all the classified images. This low error rate suggests a high level of accuracy in the classification results.

The feature space plot (Table 3), which visualises the distribution of the training samples in the classification feature space and shows a close-to-perfect linear correlation in all cases, further confirming the reliability of the classification.

Specifically, for the change detection analysis from February 1987 to February 2021, the OOB error rate was 0.02%, indicating a very high level of accuracy in detecting land cover changes over this period. The feature space plot for this analysis also exhibited a near-perfect linear correlation.

The confusion matrix showed that the model achieved 100% accuracy for class 3 and 99% accuracy for class 4, with only minor misclassifications for classes 1 and 2.

The model's efficiency, calculated as the ratio of correctly classified samples to the total number of samples, was above 90% for all the date pairs studied, with an efficiency of 98.4% for the February 1987–2021 change detection analysis.

The OOB error rate estimate of 0.02 (2%) for the 1987–2021 change detection model indicates a relatively low error rate, suggesting good predictive performance.

The feature space plot suggests that the model's performance generally improves as the layer 2 value increases, with the higher classes (6 and 7) achieving the best results.

The accuracy assessment results demonstrate that the Random Forest classification model used in this study had a very high level of accuracy, with low error rates, high efficiency, and strong performance across different land cover classes. This provides confidence in the reliability of the vegetation cover change detection analysis presented in the previous sections.

The quantification of uncertainty for the 1986DEC-2020DEC period, the mapped vegetation area was 40,000 ha. After adjustment for classification accuracy, the estimated area was 39,200 ha (95% CI: 38,102–40,298 ha). Similar calculations were performed for all other periods.

Discussion

Magnitude and pattern of vegetation loss

The Afram Plains have experienced a dramatic 75% reduction in vegetation cover from 1987 to 2021, with only 5% of the original cover remaining. This acute loss is among the most severe documented in Ghana's forest-savannah transition zone and is consistent with findings from other studies in the region, which highlight rapid land cover loss driven by human activities such as farming, charcoal production, and land use change[18,49–51]. Appiah[16], reported predominantly open forests with less than 2% cover in the Afram plains.Dahan et al.[49], conducted a multi-district study including the Afram Plains and Sene districts, using remote sensing (2001–2021) to show high rates of land cover loss in both Guinea-savannah and forest-savannah mosaic zones, with losses outpacing gains and stability, especially in the last decade. This corroborates this finding by providing specific insight into the degradation intensity and temporal trend of the Afram Plains. The identification of over 2,200 charcoal scars and the annual depletion of nearly 400,000 trees in our recent paper[18] provides a direct, activity-based context for the NDVI-detected vegetation loss in this study.

The severe reduction in vegetation cover has critical implications for ecosystem services. Vegetation is fundamental for water regulation, soil protection, and biodiversity support. Studies show that loss of green cover directly reduces the capacity of ecosystems to provide these services, leading to increased soil degradation, reduced water retention, and loss of biodiversity[52–54]. When vegetation cover falls below critical thresholds, as seen in the Afram Plains, the ability of the landscape to maintain essential ecosystem functions is severely compromised[52,54,55].

The Afram Plains, as part of the forest-savannah transition, play a crucial role in maintaining regional ecological balance. The observed vegetation loss could trigger cascading effects, such as the southward expansion of the savannah, further biodiversity loss, and altered regional climate patterns[54,56]. Additionally, the loss of this significant carbon sink undermines Ghana's climate change mitigation efforts and its commitments under international environmental agreements[55,56]. Global and African studies confirm that such ecosystem degradation, when combined with climate change, can push systems past tipping points, leading to potentially irreversible shifts in ecosystem structure and function[54,56].

Temporal dynamics of deforestation

The temporal analysis of deforestation in the Afram Plains reveals a complex pattern, with both persistent long-term vegetation loss and dramatic short-term spikes. While the average long-term deforestation rate from 1986 to 2020 was 669 hectares per year, the period of 2017–2018, which saw an acute surge to 54,953 hectares per year linked to an acute national energy crisis, suggests that while persistent baseline deforestation continues, acute socio-economic pressures can trigger dramatic spikes in vegetation loss.

The observed post-deforestation regeneration highlights a degree of ecosystem resilience, but the inability of regrowth to fully offset losses points to a weakening system. This aligns with broader findings that, while forest structure and fragmentation can be effectively monitored with remote sensing, indicators of true regeneration and compositional recovery are more challenging to capture and often lag behind structural changes[57]. As such, persistent and acute deforestation events may push the ecosystem past a threshold where natural recovery is no longer viable.

Reforestation and ecosystem resilience

Our analysis reveals a dynamic landscape in the Afram Plains, where deforestation and natural regeneration processes occur simultaneously. While the median annual deforestation rate is 4,728 hectares, concurrent natural regeneration occurs at a median rate of 2,837 hectares per year. This pattern, especially the exceptional regrowth of 54,070 hectares during 2015–2017, demonstrates the ecosystem's inherent resilience and capacity for recovery following disturbance.

Despite these signs of resilience, the rate of natural regeneration remains insufficient to fully offset ongoing deforestation, with regrowth representing only about 60% of annual losses. This imbalance highlights the challenge of achieving net positive vegetation recovery under current land use and disturbance regimes. The cyclical pattern of loss and recovery, with more dramatic regeneration in shorter periods, suggests that while the ecosystem can respond to reduced pressure, sustained and large-scale recovery is difficult without targeted intervention.

The observed resilience offers hope for restoration, but it also underscores the need for proactive management to enhance and accelerate natural regeneration. Evidence from global and regional studies emphasizes that restoration strategies especially those prioritizing natural regeneration and mixed-species reforestation are most effective for recovering ecosystem services, biodiversity, and carbon storage[58,59]. Mixed plantations, for example, have been shown to enhance ecosystem connectivity and resilience, supporting a broader range of ecological functions compared to monocultures[60].

Charcoal production as the primary driver of Afram Plains deforestation

This dramatic spike in annual forest loss of 54,953 hectares which coincides with Ghana's nationwide energy emergency characterized by severe shortages of liquefied petroleum gas (LPG) between 2017 and 2018 which significantly disrupted household energy access, particularly in urban centers, and triggered a widespread reversion to charcoal as the primary cooking fuel (GhanaWeb[61], underscores a direct and compelling link between energy insecurity and deforestation.

Despite national efforts to promote LPG as a cleaner alternative to reduce deforestation, persistent supply chain issues and unreliable distribution systems have consistently undermined these goals. As LPG became both scarce and expensive, urban demand for charcoal surged, placing immense pressure on the woodlands of the Afram Plains. Research by Ablo et al.[62] and ecosystems Afari-Asiedu et al.[63] shows that disruptions in LPG access in Ghana's savannah regions lead households to revert to traditional biomass fuels, particularly charcoal. This behavioral shift significantly intensifies charcoal production, which is closely associated with vegetation loss and ecological degradation. The 2017–2018 deforestation peak reflects a broader pattern observed across sub-Saharan Africa, where energy crises often result in increased reliance on woodfuels (Food and Agriculture Organization[64].

Methodological implications

Combining remote sensing, NDVI time series, and advanced machine learning algorithms such as Random Forest significantly improves the accuracy of land cover change detection, especially in complex or seasonally variable environments. Studies confirm that Random Forest outperforms other classifiers like SVM and ANN, achieving high overall accuracy (e.g., 94.88% and Kappa 0.94) in crop and land cover mapping using NDVI time series from high-resolution images[65]. Integrating NDVI with machine learning not only enhances the detection of subtle vegetation changes but also addresses challenges posed by low-resolution data and complex spectral signatures.

The methodology employed in this study integrating supervised classification, NDVI estimation, and Random Forest demonstrated exceptional accuracy, with Out-of-Bag (OOB) error rates below 0.04% and overall efficiency exceeding 98% which surpasses the common threshold of that aligns with best practices in recent literature[65,66]. The careful selection of training samples, the ability of Random Forest to handle complex data, and the use of multiple temporal images all contribute to this high performance.

The use of NDVI as a key input, especially when combined with machine learning models, consistently yields high classification and prediction accuracy. For example, hybrid approaches using NDVI and machine learning have achieved accuracy rates up to 96% in land cover classification tasks[65,66]. This demonstrated effectiveness of integrated remote sensing and machine learning approaches provides a robust framework for land cover monitoring in data-limited regions.

Policy implications

Our results highlight the urgent need for integrated policy responses that address both environmental conservation and energy security. There is an urgent need for targeted interventions during periods of acute socio-economic stress, as well as sustained efforts to support ecosystem resilience. Effective interventions could include: Expanding access to affordable, sustainable energy (e.g., LPG, solar) to reduce reliance on charcoal and woodfuel. Improved enforcement of land use regulations and community-based resource management can help curb illegal tree harvesting. Incentivizing reforestation and agroforestry in degraded areas can enhance ecosystem resilience and livelihoods. Policymakers should recognize the direct impact of national energy policy on land cover dynamics and prioritize cross-sectoral strategies that simultaneously address energy needs and forest conservation. Future research should leverage higher-resolution imagery, expand field validation, and incorporate socio-economic data to further elucidate the drivers and consequences of vegetation change in the Afram Plains. To maximize the benefits of reforestation and ecosystem resilience, conservation strategies should prioritize protection of existing forests and promote natural regeneration where possible. And also monitor and adapt management practices to respond to changing disturbance regimes and ecological feedbacks.

Limitations of the study

Some limitations include, the limited availability of cloud-free satellite imagery, resulting in temporal gaps. While Landsat imagery provides valuable long-term coverage, its moderate resolution may not detect small-scale vegetation changes or degradation beneath canopy cover. Although ground truth data were used to refine classification, logistical constraints limited the extent of field validation. While strong associations between energy crises and deforestation were observed, the study design does not allow for definitive attribution of causality.

Conclusions

This study provides robust evidence that charcoal production is the dominant driver of vegetation loss in the Afram Plains, Ghana, with profound implications for both local ecosystems and national energy policy. Over the 34-year period analyzed, the region experienced a dramatic 75% reduction in vegetation cover, far exceeding national and global deforestation rates. This rapid decline is closely linked to energy crises, particularly the 2017–2018 LPG shortage, which triggered unprecedented spikes in charcoal demand and deforestation.

Despite the ecosystem's demonstrated resilience evidenced by notable natural regeneration rates this capacity is increasingly insufficient to offset ongoing losses. The median annual regeneration of 2,837 hectares, while significant, cannot keep pace with the median deforestation rate of 4,728 hectares per year. As vegetation cover diminishes, the risk of crossing ecological thresholds and losing critical ecosystem services intensifies.

Our findings challenge the prevailing national narrative that attributes deforestation in Ghana primarily to agricultural expansion and logging[67]. Instead, we highlight the central role of charcoal production, particularly during periods of energy insecurity, as a dominant driver of vegetation loss in the Afram Plains. This pattern mirrors broader trends across sub-Saharan Africa, where urban demand for charcoal often driven by unreliable access to modern energy sources has emerged as a major cause of forest degradation and deforestation, frequently independent of agricultural drivers[68–71]. The Afram Plains, in particular, exhibit deforestation rates that surpass national averages, underscoring the disproportionate impact of energy-driven charcoal production in non-cocoa regions. These findings suggest that national forest assessments may underestimate the role of energy insecurity in driving deforestation. As such, conservation policies must be tailored to local realities, integrating energy security with forest management. Strengthening access to sustainable energy alternatives and regulating charcoal production are therefore critical for effective forest conservation in the Afram Plains and similar landscapes.

Methodologically, the integration of supervised classification, NDVI estimation, and Random Forest algorithms yielded exceptionally high accuracy, demonstrating the value of advanced remote sensing and machine learning techniques for monitoring land cover change in resource-constrained settings.

The implications are clear. The effective conservation in the Afram Plains requires integrated strategies that address both forest management and national energy security. Policy interventions should prioritize regulated charcoal production, promote alternative energy sources, and support active restoration programs. Sustainable charcoal production systems—such as those involving improved carbonization efficiency, communal management, and demand reduction—offer promising pathways for balancing energy needs with ecosystem conservation[72,73]; van 't Veen et al.[74]; Van '[75].

The Afram Plains' experience serves as a warning of the environmental risks posed by unregulated resource extraction, but also as a source of hope. With evidence-based, adaptive management and policy reform, it is possible to halt and reverse ecosystem degradation, ensuring the long-term sustainability of both local livelihoods and natural resources. Future research should focus on identifying ecological thresholds, refining restoration techniques, and developing sustainable energy models that support both people and the environment.

In preparing this work, the author(s) used Hugging Face and Claude AI to improve the language and readability. The author(s) have reviewed and edited the content and take(s) full responsibility for the publication's content.

Data availability

Data is available on request. Contact the corresponding author.

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Author contributions

TA made substantial contributions to the conception or design of the work or the acquisition, analysis, or interpretation of data, drafted the work, and agreed to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved. AM supervised the work and approved the version to be published. JA supervised the work and approved the version to be published. FD supervised the work and approved the version to be published. PO supervised the work and approved the version to be published.

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Declarations

Competing interests

The authors declare no competing interests.

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